

Beyond SMA and EMA: Demonstrating the Strength of the Kalman Filter for Embedded Sensor Smoothing

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Abstract. The study offers an overall comparison between the simple moving average (SMA) algorithm, the exponential moving average (EMA) algorithm, and the Kalman filter algorithm used as a filter to smooth out the temperatures. The overall comparison has been done using the simulated and actual measures taken by the DHT22 sensors when implemented on resource-limited microcontrollers used by the Internet of Things (IoT) environment. The results reveal that the MAE value is 0.3592, the measure of the level of instability is 8.66%, and the value of robustness is 88.19% when the Kalman filter algorithm is used.

Introduction

In IoT scenarios, memory and processing capabilities are very limited, especially for entry-level microcontrollers like the ATmega328P on the Arduino UNO, which only has about 2 KB of SRAM at a clock speed of 16 MHz, thus restricting the complexity of the filters that can be implemented[1], [2]. In addition, temperature, humidity, and gas sensors are well known to provide data that can often be marred by the presence of noise from the electrical signal as well as an intermittent anomaly[3], [4]. Therefore, it is necessary to implement efficient and simple denoising methods to avoid measurement errors during monitoring and control. Traditional smoothing filters involve compromises, such as the simple moving average (SMA)[5], [6]. The exponential moving average (EMA), on the other hand, offers a faster response with a memory complexity of $O(1)$, but at the cost of a fixed smoothing parameter α , which inevitably implies a trade-off between responsiveness and noise amplification[7], [8]. The Kalman filter offers a probabilistic model for dynamically approximating the optimal state estimate within a Gaussian framework, but it is excessively dependent on the relevance of the process and measurement error covariances (Q and R), even though it produces suboptimal estimates in the presence of non-Gaussian errors or outliers. Within this limited context, this paper proposes a comprehensive comparative approach to the SMA, EMA, and Kalman filters, applied to synthetic DHT22 temperature data and real-world examples[9], [10], [11]. It also presents a set of numerical tests enabling a quantitative comparison of the filters using error measures such as mean error, root mean square error (RMSE), robustness, and resistance to outliers. These tests demonstrate how Kalman filters represent an advantageous compromise in the context of the embedded Internet of Things (IoT), while



presenting an adaptive strategy for a real-time dynamic smoothing model, capable of automatically learning its thresholds in order to overcome the limitations of fixed-parameter filters.

Filtering Methods: Principles and Approaches

In this section, we describe three classical signal smoothing techniques that balance computational efficiency with noise suppression performance. Each method represents a different trade-off between memory usage, computational complexity, and noise-responsiveness, making them well-suited for evaluation on severely resource-constrained embedded platforms[3], [12], [13].

Simple Moving Average (SMA). The Simple Moving Average computes the output as the arithmetic mean of the N most recent input samples as on Eq. 1:

$$y_t = \frac{1}{N} \sum_{i=0}^{N-1} x_{t-i} \quad (1)$$

The Simple Moving Average (SMA) delivers a very stable output in steady-state regions and strongly attenuates high-frequency noise, but it introduces tracking delays proportional to the window size and requires storing N past samples, leading to $O(N)$ memory and, in naive form, $O(N)$ computations per update, although a circular-buffer implementation can reduce the operations to $O(1)$. Consequently, SMA is well suited to slow-varying signals where stability is prioritized over responsiveness, but it is poorly adapted to signals with rapid environmental changes or frequent step transitions[14], [15].

Exponential Moving Average (EMA). The Exponential Moving Average uses a weighted combination of the current sample and the previous filtered value as on Eq. 2:

$$y_t = \alpha x_t + (1 - \alpha)y_{t-1} \quad (2)$$

where:

- $0 < \alpha \leq 1$ is the smoothing factor
- High $\alpha \rightarrow$ fast response (more weight on current measurement)
- Low $\alpha \rightarrow$ strong smoothing (more weight on history)

The Exponential Moving Average (EMA) adapts quickly with constant $O(1)$ memory and computation, which makes it attractive for microcontrollers, but its fixed smoothing factor α forces a trade-off where higher α improves responsiveness while amplifying noise and lower α enhances smoothing while reducing responsiveness, so α must be manually tuned for each application[16], [17], [18].

Kalman Filter. The Kalman filter employs a prediction-correction framework to provide a probabilistic estimate of the underlying signal and its estimation uncertainty. For scalar IoT sensors (one-dimensional case), the filter operates in two steps as on Eq. 3 and Eq. 4:

$$\hat{x}_{t|t-1} = \hat{x}_{t-1} \quad (3)$$

$$P_{t|t-1} = P_{t-1} + Q \quad (4)$$

Correction Step

$$K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R} \quad (5)$$

$$\hat{x}_t = \hat{x}_{t|t-1} + K_t(z_t - \hat{x}_{t|t-1}) \quad (6)$$

$$P_t = (1 - K_t)P_{t|t-1} \quad (7)$$

where:

- $\hat{x}_t$ = filtered estimate at time t
- P_t = estimation error covariance (uncertainty)
- Q = process noise covariance
- R = measurement noise covariance
- K_t = Kalman gain (optimal blending factor)

The Kalman filter is theoretically optimal under Gaussian noise and updates with constant $O(1)$ cost while automatically adapting its gain, but its performance is highly sensitive to the tuning of Q and R and can degrade in the presence of non-Gaussian noise or strong outliers[19], [20].

Experimental Results

Experimental Setup and Dataset. To this end, a 100-second synthetic temperature signal was created, incorporating reference values through the addition of constant levels, noisy segments, slowly varying levels, and outliers introduced by Gaussian white noise. In addition, real-world DHT22 temperature data was collected using an Arduino UNO board and a low-cost temperature sensor with a typical accuracy of ± 0.5 °C. This allows for the integration of real-world phenomena such as thermal drift, correlated noise, and signal loss, which are difficult to model with synthetic data.

In our experiment, the filter parameters are chosen for ease of comparison and represent commonly used tuning parameters. The SMA window length, N , is set to the minimum required noise reduction with minimal attenuation of process variation, i.e., $N = 5$ units. The EMA smoothing parameter, α , is set to 0.3 to minimize MAE values in simulations using the generated data, which corresponds to a "fast and noisy" processing method. Finally, the Q and R parameters of the Kalman filter are determined using observations of the natural variability of the DHT22 readings and noise as a guide for parameter tuning and are set to $Q = 0.01$ and $R = 0.5$ [21], [22], [23].

Qualitative Analysis

Noise Filtering in Steady-State Regions. Fig. 1 illustrates the application of the filtering process to the constant temperature area with high noise. The SMA Filter removes most of the noise and provides the most accurate results after a small delay, while the EMA Filter is still sensitive to the tiniest changes and thus shows noise. The Kalman Filter should remove noise significantly while preserving responsiveness, and that is the case.

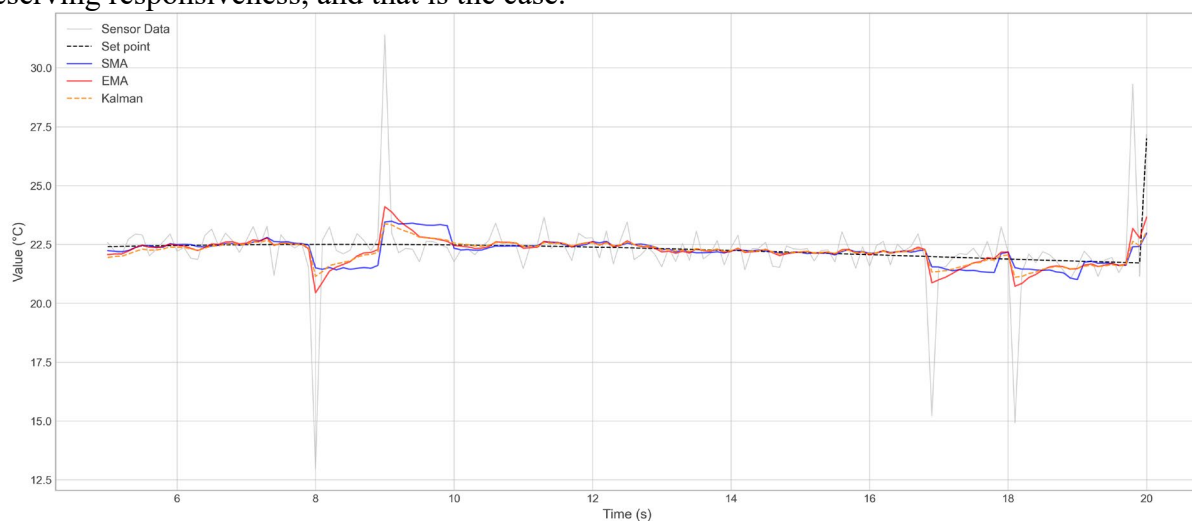


Fig. 1: Noise Filtering Zoom

Under constant-temperature conditions, this zoomed view compares the noise-removal performance of SMA, EMA, and the Kalman filter on the same sensor signal. The SMA strongly smooths the noise but flattens the signal and introduces a noticeable delay, while the EMA remains more responsive but leaves visible residual jitter when using a fixed $\alpha = 0.3$. The Kalman filter achieves a better compromise by significantly attenuating high-frequency noise, preserving sensitivity to small variations, and reaching a steady-state stability score of 8.66%, which positions it between SMA and EMA in terms of responsiveness.

Response and Transient Tracking. Fig. 2 depicts two plateaus in the signal input. Although the SMA filter traces the slow and continuous filtering process along the averaging window, the EMA filter responds to the plateaus after a slight delay but with overshooting and settling around the new value. The Kalman filter responds to the plateaus gradually without overshooting, providing a balance between response speed and stability.

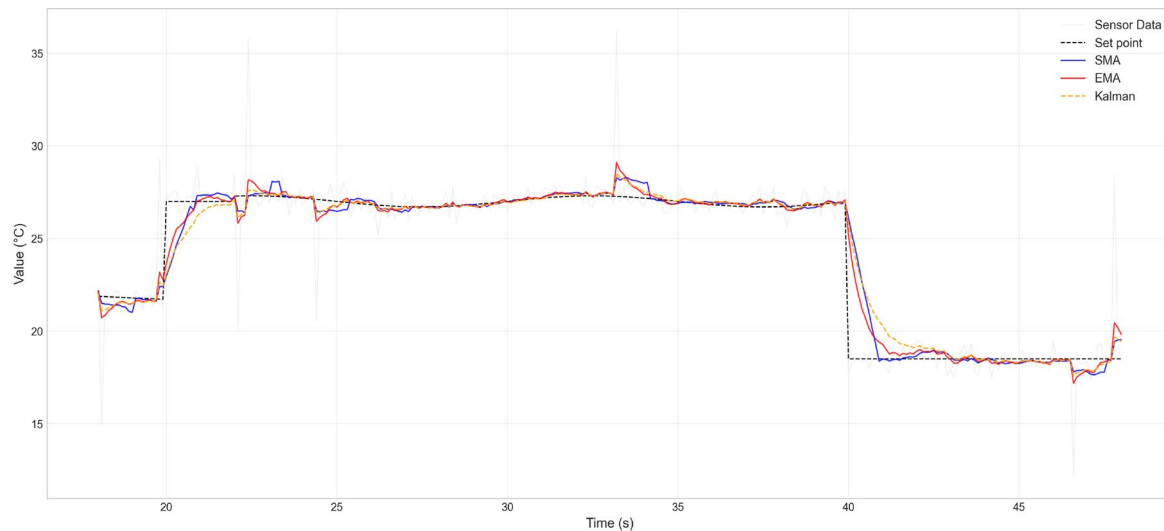


Fig. 2: Step Response and Transient Behavior (Zoom View).

Two independent step changes highlight the dynamic behavior of the three filters: the SMA responds with a slow ramp over 2–3 seconds, trading reactivity for strong noise rejection, while the EMA reacts quickly but overshoots. Le facteur de quantification mise en évidence l'importance du fait que chacune des filtres traditionnelles se situe dans une région uniquement à elle au compromis erreurs, robustesse et stabilité, alors que le kalman offre le meilleur compromis. The Kalman filter tracks the steps smoothly and rapidly without overshoot, because its adaptive gain based on covariance P continuously balances prediction and measurement, delivering both fast transients and a clean steady-state output compared with the fixed window of the SMA and fixed gain of the EMA.

Signal Behavior. Fig. 3 complete simulated signals of 100s with stable phases, sudden transitions, small changes, and noise peaks. In the entire period, the simple moving average correctly detects the stable phases, with constant lag in sudden changes. On the other hand, the exponential moving average filter reacts rapidly per change, with largely intensified noise in almost stable phases. The Kalman filter manages the trend correctly without considerable oscillation, despite the existence of noise.

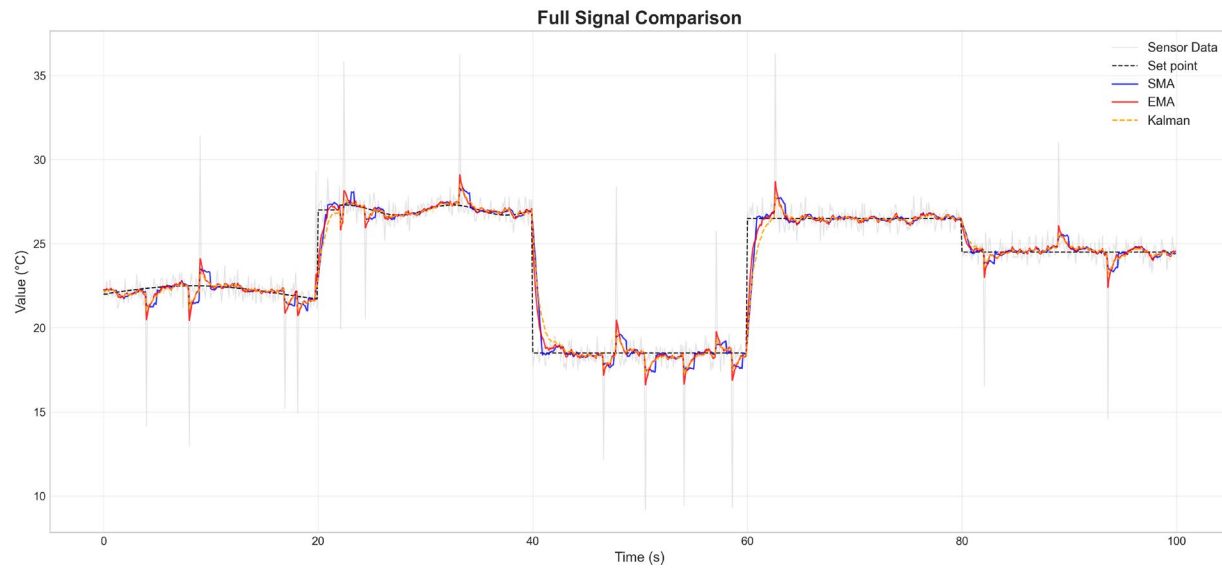


Fig. 3: Full Signal Comparison Over 100-Second Duration.

On the complete artificial data set, which includes constant zones, ramps, discontinuities, and outliers, the three filters behave very differently. The SMA is efficient in stationary regions but reacts with a noticeable delay during transitions, while the EMA responds quickly to changes yet amplifies noise in quasi-steady regimes, causing oscillations and instability. By contrast, the Kalman filter closely follows the true signal with only a small delay, suppresses noise and abnormal peaks, and remains consistent across all operating conditions, adapting well to real-world variations without special tuning.

Quantitative Evaluation. Quantitative measurements based on the synthetic signal obtained from the known reference signal are presented in Table 1. In addition to error measurements, two aggregate measurements were calculated:

- **Stability (%):** This is an expression of the average suppression of high-frequency noise energy when in a state of steady state, as a fraction of the reduction in standard deviation from the raw signal to the filter output.
- **Robustness (%):** This measurement represents the filter's ability to eliminate deviations caused by the presence of outliers. It is expressed as a percentage and represents the proportion of deviations eliminated by the filter relative to the outliers.

Table 1. Performance Comparison of SMA, EMA, and Kalman Filter on Synthetic Temperature Dataset

Method	Stability (%)	Robustness (%)	MAE	RMSE
SMA	6.88	91.12	0.3548	0.8120
EMA	7.31	81.37	0.3320	0.6922
Kalman	8.66	88.19	0.3592	0.8139

Balanced filtering performance for embedded IoT sensors

The quantification factor highlights that each traditional filter occupies a unique position regarding the trade-off between error, robustness, and stability, whereas the Kalman filter offers the best overall compromise. The EMA offers the lowest errors (\$MAE=0.3320\$ and \$RMSE=0.6922\$), indicating excellent sensitivity to signal variations, but this comes with the lowest robustness (81.37%), meaning marked sensitivity to outliers and noise in supposedly stable regions. The opposite is true for the SMA, which has maximum robustness (91.12%), meaning excellent ability to reject outliers and suppress noise, but low stability sensitivity (6.88), as well as a higher MAE

indicating marked delays in transition mode. The Kalman filter benefits from both; although it has an MAE slightly higher than the EMA (0.3592), it offers better stability sensitivity (8.66) and moderate robustness (88.19%). This higher MAE represents an acceptable compromise for exceptional smoothing capacity in stable modes and low sensitivity to perturbations, making it the most consistent choice for IoT applications.

Conclusion and Perspectives

This paper systematically compares three traditional filtering algorithms: Simple Moving Average Filter (SMAC), Exponential Moving Average Filter (EMA), and Kalman Filter (KAF), and uses simulated and actual measurement data from the DHT22 sensor to perform noise reduction on temperature data. This allows us to evaluate the efficiency and limitations of each algorithm in denoising before outputting temperature readings. Simple Moving Average Filter performs well in noise reduction, which has a robustness of 91.12%, but the filtering speed is slow and the output of filtering has a time delay; thus, the method could be suitable for slow monitoring processes where this kind of time delay is acceptable. The Exponential Moving Average Filter is fast, which has a mean absolute error of 0.3320, but it cannot effectively avoid the impact of the largest outlier, with only a robustness of 81.37%, and the noise reduction performance is poor. Hence, this algorithm would perform best in an application with an extremely high request for response speed. Kalman filter, as the algorithm performing the best, demonstrates stability, the response value is 8.66, robustness of the generated data is 88.19%, and moderate in response speed, mean absolute error 0.3592, which thus proves the academic assertion about the reliability of the Kalman filter algorithm in comparison with other traditional embedded sensor system algorithms. That said, this indicates that, for noisy IoT applications, the Kalman filter algorithm will be more efficient compared to the other two algorithms. Due to the characteristics of its probability, the algorithm can adjust dynamically; hence, in the IoT environment with noises, to ensure the speed and robustness of the algorithm, the Kalman filter algorithm is the most efficient among the three algorithms. Only the sensitivity of the parameters to Gaussian noise and large outliers needs further adjustments. In the future, we will focus on eliminating the above limitations by implementing a dynamic algorithm, such as automatic threshold adjustment, outlier removal, and dynamic mode switching. Then, the algorithm will be put into operation and physically tested through chip/PCB implementation to meet the needs of real-world IoT application environments.

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